INCH-POUND

MIL-PRF-19500/552J
w/AMENDMENT 1
14 June 2023
SUPERSEDING
MIL-PRF-19500/552J
5 July 2018

The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 October 2023.

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, UNIPOLAR TRANSIENT VOLTAGE SUPPRESSOR, TYPES 1N6469 THROUGH 1N6476, THROUGH-HOLE AND SURFACE MOUNT PACKAGES, QUALITY LEVELS JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 1,500-watt silicon, transient voltage suppressor diodes. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outlines and die topography. The device packages for the encapsulated device types are as follows: Axial lead in accordance with figure 1, square end-cap surface mount version US in accordance with figure 2, and an one round end-cap and one square end-cap surface mount version URS in accordance with figure 3.

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings (see 3.6 herein) and as follows:

a. P_R = 3 W (derate 20 mW/°C above T_A = +25°C) (see figure 4).

b. Derate at 100 mW/°C above T_{EC} = +145°C for surface mount devices (see figure 5).

c. P_{PP} = 1,500 W at T_P = 1 ms (see figure 6).

d. I_{FSM} = 130 A(pk). T_P = 8.33 ms (T_A = +25°C).

e. -55°C ≤ T_J ≤ +175°C; -55°C ≤ T_{STG} ≤ +175°C (ambient).

1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in maximum test ratings (see 3.6 herein).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

FSC 5961

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The JAN certification mark and quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductor of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductor devices covered by this specification sheet are as follows: "6469", "6470", "6471", "6472", "6473", "6474", "6475", and " 6476".

1.5.3 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

	A blank suffix symbol indicates a through-hole mount axial package (see figure 1).
US	Indicates a surface mount package with square end-caps (see figure 2).
URS	Indicates a surface mount package with one round end-cap and one square end-cap (see figure 3).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

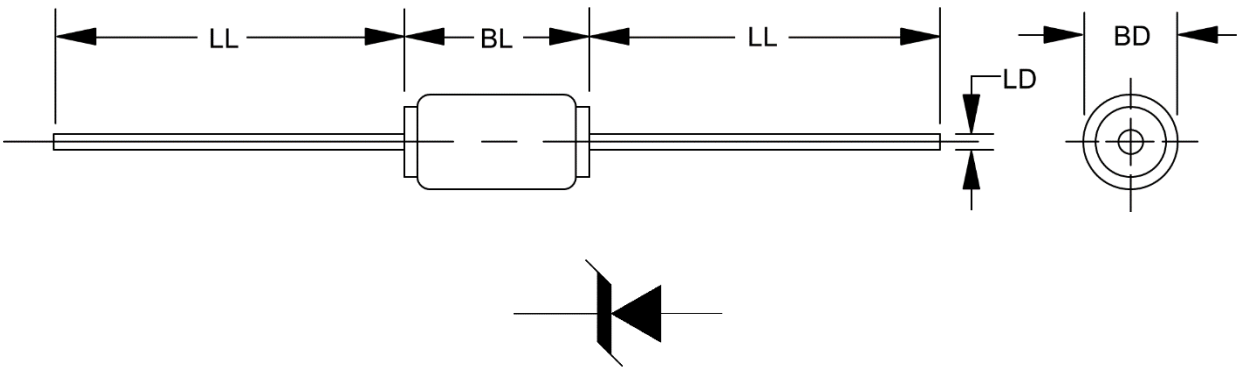
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.



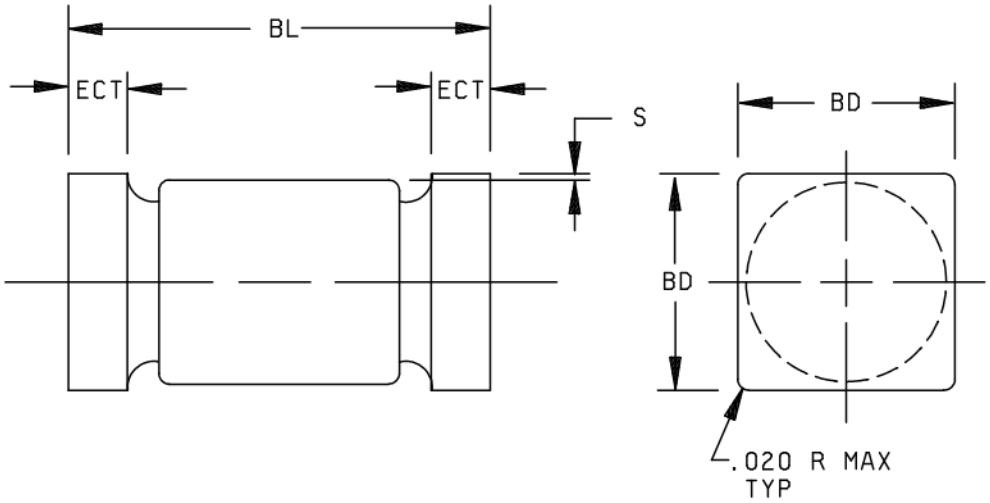
SCHEMATIC SYMBOL

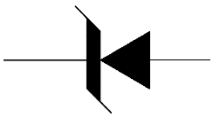
Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BD	.150	.185	3.81	4.70	2
BL	.150	.375	3.81	9.53	2
LD	.037	.042	0.94	1.07	3
LL	.900	1.300	22.86	33.02	
LU		.050		1.27	4

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimension BD shall be measured at the largest diameter. Package contour optional within dimensions BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of dimension BD.
3. Dimensions are pre-solder dip.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities other than heat slugs.

FIGURE 1. Physical dimensions for round body, axial leaded devices (1N6469 through 1N6476).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.





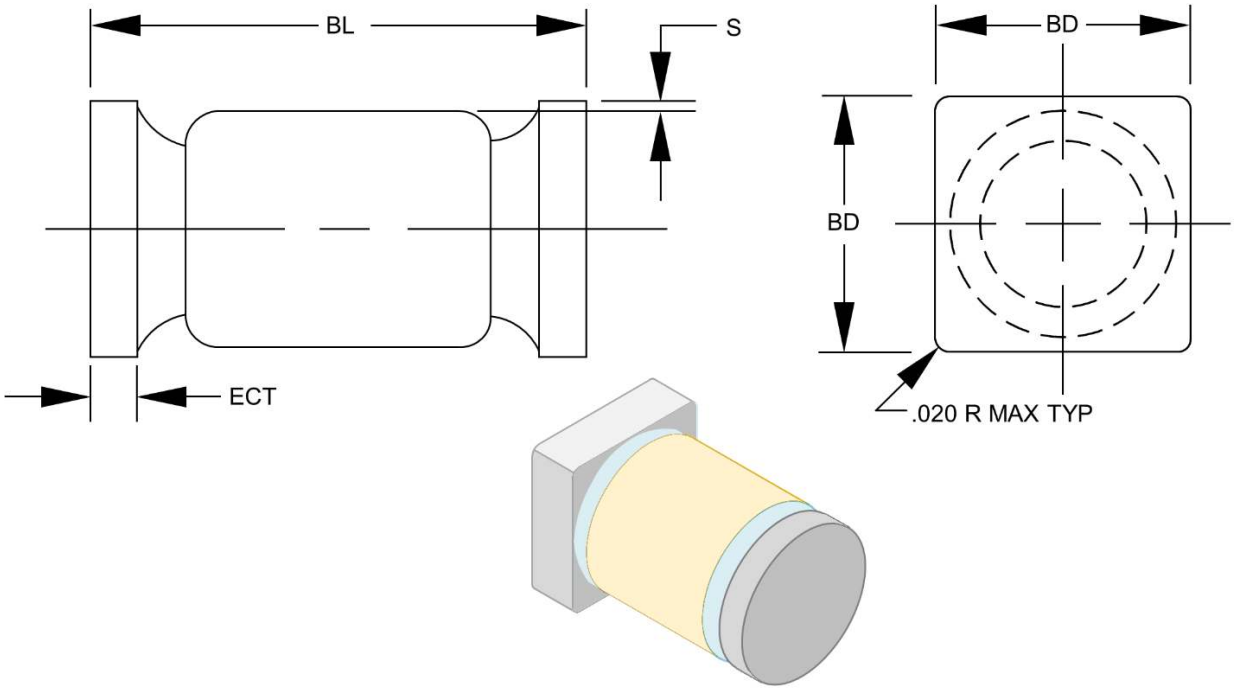
SCHEMATIC SYMBOL

Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BD	.183	.202	4.65	5.13	2
BL	.205	.245	5.21	6.22	2
ECT	.019	.028	0.48	0.71	3
S	.003		0.08		

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension S is optional however the glass body diameter shall not exceed endcap diameter.

FIGURE 2. Physical dimensions for round body, square end-cap surface mount (US) packages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.



Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BD	.183	.202	4.65	5.13	2
BL	.205	.245	5.21	6.22	2
ECT	.019	.028	0.48	0.71	3
S	.003		0.08		

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. One endcap shall be square and the other endcap shall be round.
4. See figures 1 or 2 for schematic symbol.

FIGURE 3. Physical dimensions for round and square end-cap surface mount (URS) package.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

V_{CF} Forward clamping voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, and 3 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Metallurgically bonded construction. These devices shall be constructed to comply with the requirements of a category I metallurgical bond of MIL-PRF-19500, and herein.

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500. All device marking (including all marking which is omitted on the device), except for polarity and serial numbers, shall appear on the unit package used as the initial protection for delivery.

3.5.1 Device marking abbreviations. The prefixes JAN, JANTX, and JANTXV may be abbreviated as J, JX, and JV respectively. For example: The PIN JANTXV1N6469URS may be reduced to JV6469. Color coding shall not be permitted for part identification.

3.5.2 Marking of US and URS versions. For US and URS versions only, all marking may be omitted from the device except for the cathode marking. US and URS devices shall be marked with a cathode band as a minimum; or a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used. At the option of the manufacturer, URS devices may include laser marking on an end-cap, to include the PIN and lot date code for all levels.

3.5.3 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UR suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used. For URS surface mount parts only, cathode shall be connected to the round end-cap.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and tables I and II herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I subgroup 2.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table IV tests, the tests specified in table IV herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANTX and JANTXV levels
9	Not applicable
11	I_D and V_{BR}
12	See 4.3.1
13	$\Delta I_D \leq 50$ percent of initial reading or 20 percent of column 5 of table II herein, whichever is greater; $\Delta V_{(BR)} \leq \pm 2$ percent of initial reading.

4.3.1 Power burn-in and steady-state operation life test conditions. This test shall be conducted with the devices subjected to test conditions in the following order of events and conditions:

a. Pulse in accordance with 4.5.2.a, 20 times (screening and group B operation life test) and 100 times (group C) at T_A = +25°C.

b. Read I_D and V_{BR} at T_A = +25°C, remove defective devices, and record the number of failures.

c. Apply the working peak reverse voltage (V_{WM}) (column 4 of table II herein) at an ambient temperature of T_A = +125°C, as follows. End-point measurements shall be performed within 24 hours.

(1) 96 hours minimum (JANTX and JANTXV) for the screening test.

(2) 340 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) 1,000 hours for group C steady-state operation life test.

d. Read and record I_D and V_{BR} at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of initial value or 20 percent of column 5, table II herein, whichever is greater, or ΔV_{BR} > 2 percent (±5 percent for steady-state operation life) of initial value shall be considered defective. For steady-state operation life, I_D limit (maximum) shall be two times the table I, subgroup 2 limit. Remove defective devices and record the number of failures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB of MIL-PRF-19500. Delta measurements shall be in accordance with the applicable steps of table III herein.

Subgroup	Method	Conditions
B2	1051	-55°C to +175°C, 25 cycles n = 22, c = 0.
B3	1026	See 4.3.1.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with the applicable steps of table III herein.

Subgroup	Method	Conditions
C2	2036	Test condition A, 20 pounds, t = 15 seconds, ±3 seconds.
C2	2036	Axial leaded devices only. The details for terminal strength shall be as follows: Tension (condition A). 20 pounds (9.1 kg), t = 15 seconds, ±3 seconds. Lead fatigue (condition E). Applicable for all types, 2 pounds (0.91 kg).
C2	2038	US and URS devices only. The details for end-cap bond integrity shall be as follows: Tab pull (condition B). 20 pounds (9.1 kg), t = 15 seconds, ±3 seconds. Suitable fixtures may be used to pull the end-caps in a manner which does not aid construction. Reference to axial lead may be interpreted as end-cap with fixtures used for mounting.
C5	4081	$R_{\theta JL}$ ≤ 50°C/W at L = .375 inch (9.53 mm), non-surface mount devices. $R_{\theta JEC}$ ≤ 20°C/W at L = 0 inches, surface mount devices. T_A = +25°C.
C6	1026	T_A = +125°C (see 4.3.1).
C7	1018	Not required.
C8		Maximum peak pulse current; conditions, see 4.5.2.b (20 µs pulse only) 10 pulses; 22 devices, c = 0.
C9	4071	$\propto_{V(BR)}$ in accordance with method 4071 of MIL-STD-750, I_{BR} = column 3 of table II herein, T_1 = +25°C ±3°C; T_2 = +125°C; maximum limits = column 8 of table II herein; units = %/°C; 22 devices, c = 0.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. See table III herein for delta limits when applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Group C steady-state operation life test (alternate procedure). When the group B, 340 hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.3.1 with the following exceptions: 4.3.1.d shall be repeated at the end of the 1,000 hours.

4.5.2 Maximum peak pulse current (I_{PP}). The peak currents specified in column 7 of table II shall be applied in the reverse direction while simultaneously maintaining a reverse bias voltage of not less than the applicable voltage specified in column 4 of table II herein. The clamping voltage (V_C) shall be as specified in 4.5.3. The peak current shall be applied with a current versus time waveform as follows (1 pulse per minute maximum): (See figure 7, figure 8, and figure 9 herein.)

a. Pulse current shall reach 100 percent of I_{PP} at t ≤ 10 µs and decay to 50 percent at t ≥ 1 ms for t_P ≥ 1 ms.

b. Pulse current shall reach 100 percent of I_{PP} at t ≤ 8 µs and decay to 50 percent at t ≥ 20 µs for t_P ≥ 20 µs.

4.5.3 Clamping voltage (V_C). The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage, as specified in column 6 of table II herein, shall be applicable to the 1 ms pulse of 4.5.2.a only.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage	4022	tp ≤ 300 ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table II	V_{BR1}	Column 2 of table II		V dc
Standby current	4016	Condition A or B, $V_R = V_{RWM}$, (column 4 of table II)	I_{D1}		Column 5 of table II	µA dc
Subgroup 3						
Minimum breakdown voltage	4022	tp ≤ 300 ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table II; T_A = -55°C	V_{BR2}	Column 9 of table II		V dc
High temperature operation:		T_A = 125°C				
Standby current	4016	Condition A or B, $V_R = V_{RWM}$, column 4 of table II	I_{D2}		Column 11 of table II	µA dc

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Clamping voltage maximum		See 4.5.3, t_p = 1.0 ms (see 4.5.2.a) I_{PP} = column 7 of table II	V_C		Column 6 of table II	V(pk)
Forward voltage	4011	Condition A or B, I_F = 4 A dc	V_F		1.5	V(pk)
Forward voltage	4011	Condition A or B, I_{FM} = 100 A (pk), t_P = 300 µs, duty cycle = 4 pulses per minute maximum	V_{FM}		4.8	V(pk)
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, One pulse, half sine wave 8.3 ms; I_F = 0, I_{FSM} = 130 A (pk), V_{RWM} = 0, T_A = +25°C	I_{FSM}			
Electrical measurements		Table III, steps 1 and 2				
Subgroup 7						
Forward clamping voltage		See 4.5.3, t_p = 1.0 ms (see 4.5.2.a) at I_{PP} except use forward direction current without prior bias voltage	V_{CF}		Column 10 of table II	V(pk)

1/ For sampling plan, see MIL-PRF-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7a	Col 7b	Col 8	Col 9	Col 10	Col 11
Type	V_{BR1} at I_{BR} min	I_{BR} Test current	V_{WM} Working peak reverse voltage	I_{D1} Maximum standby current	V_C Maximum clamping voltage at I_{PP}	I_{PP} Maximum peak pulse current	I_{PP} Maximum peak pulse current	α_{VBR} Maximum temperature coefficient of V_{BR}	V_{BR2} Minimum at I_{BR} T_A = -55°C	V_{CF} at I_{PP}	I_{D2} Maximum I_D current at I_D T_A = + 125ºC
						tp = 1 mS tr = 10 µS (10/1,000 µs)	tp = 20 µS tr = 8 µS (8 x 20)			Inverse polarity (max)	
	V dc	mA dc	V pk	µA dc	V pk	A pk	µA dc	%/ºC	V dc	V pk	µA dc
1N6469	5.6	50	5	1,500	9.0	167		-.03, +.045	5.4	-3.5	12,000
1N6470	6.5	50	6	1,000	11.0	137	Set P_{PP} to	+.060	6.2	-3.2	8,000
1N6471	13.6	10	12	20	22.6	66	8.5 KW	+.085	12.7	-3.8	100
1N6472	16.4	10	15	10	26.5	57	minimum	+.085	15.3	-3.8	100
1N6473	27.0	5	24	5	41.4	36.5		+.096	24.9	-3.6	100
1N6474	33.0	1	30.5	5	47.5	32		+.098	30.2	-3.6	100
1N6475	43.7	1	40.3	5	63.5	24		+.101	40.0	-3.5	100
1N6476	54.0	1	51.6	5	78.5	19		+.103	48.5	-3.4	100

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

TABLE III. Groups B, and C electrical end-point and delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Conditions		Min	Max	
1.	Standby current	4016	Condition A or B, $V_R = V_{RWM}$ (column 4 of table II)	I_{D1}		Column 5 of table II	µA dc
2.	Breakdown voltage	4022	tp ≤ 300 ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table II	V_{BR1}	Column 2 of table II		V dc
3.	Standby current	4016	Condition A or B; $V_R = V_{RWM}$ (column 4 of table II)	ΔI_D	100 percent of initial reading or 20 percent of column 5 of table II, whichever is greater.		
4.	Breakdown voltage	4022	tp ≤ 300 ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table II	ΔV_{BR1}		±5 percent of initial value	
5.	Clamping voltage		tp = 1.0 ms (see 4.5.2.a); I_{PP} = column 7 of table II	V_C	Column 6 of table II		V (pk)

1/ The electrical measurements taken during group B inspections for quality level JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1, 2, 3, and 4 (see 4.3.1.c) of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1, 2, 3, and 4 (see 4.3.1.c) of this table shall also be taken.

2/ The electrical measurements taken during group C inspections for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroups 2, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroups 6, the measurements of steps 1, 2, 3 and 4 (see 4.3.1.c) of this table shall also be taken.
   c. In addition to the measurements specified for subgroups 8, the measurements of steps 1 and 2 of this table shall also be taken.

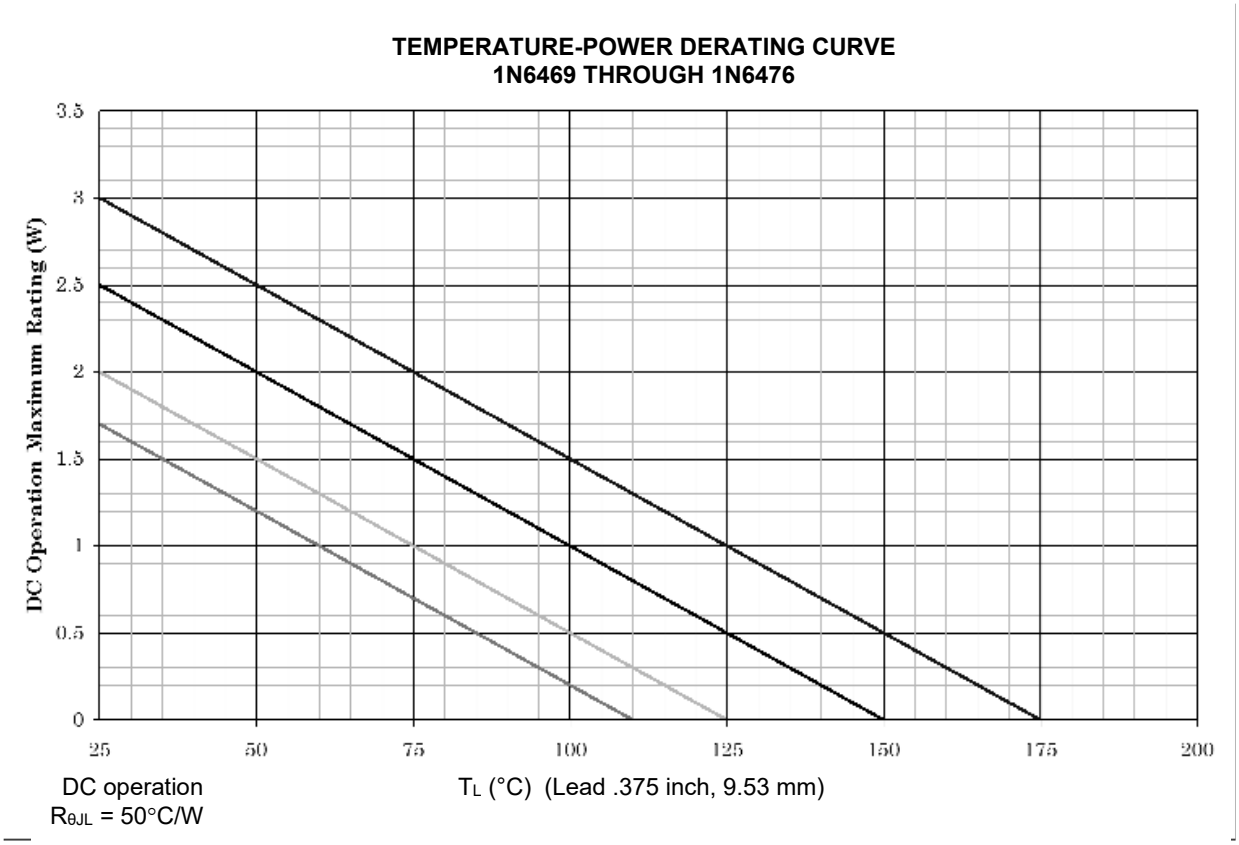
3/ Devices which exceed the table I limits for this test shall not be accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

TABLE IV. Group E inspection (all quality levels) for qualification only or re-qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			n = 45, c = 0
Temperature cycling	1051	500 cycles, condition C, -55°C to +175°C.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			n = 22, c = 0
Life test		1,000 hours. See 4.3.1. See table I, subgroup 2.	
Electrical measurements			
Subgroups 3, 4, 5, and 6			
Not applicable			
Subgroup 8			
Peak pulse current		See 4.5.2. I_{PP} shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each low and high voltage device for each structurally identical grouping. Test to failure.	
Electrical measurements		See table I, subgroup 2.	

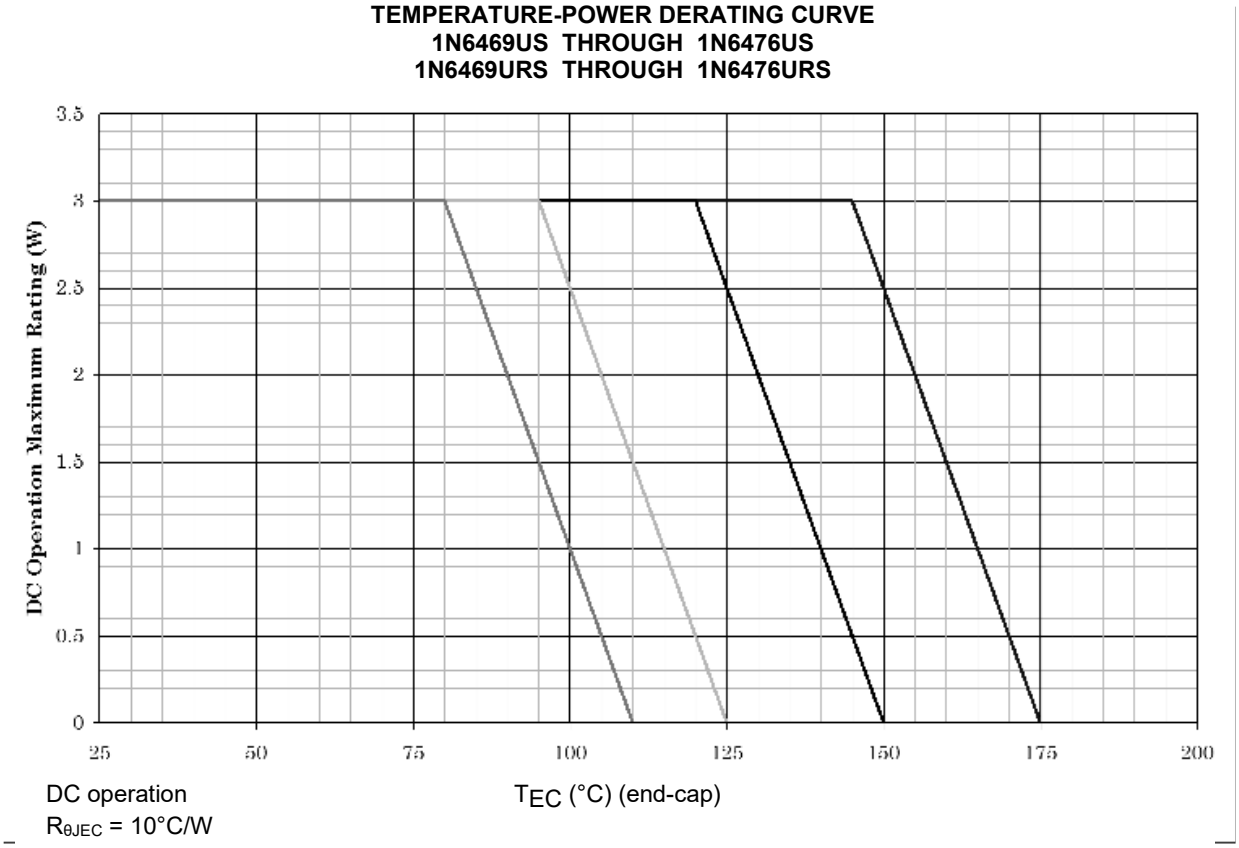
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.



NOTE: Max finish-alloy temp = 175°C

FIGURE 4 Temperature-power derating for 1N6469 through 1N6476.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.



NOTE: Max finish-alloy temp = 175°C

FIGURE 5 Temperature-power derating for 1N6469US through 1N6476US and 1N6469URS through 1N6476URS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

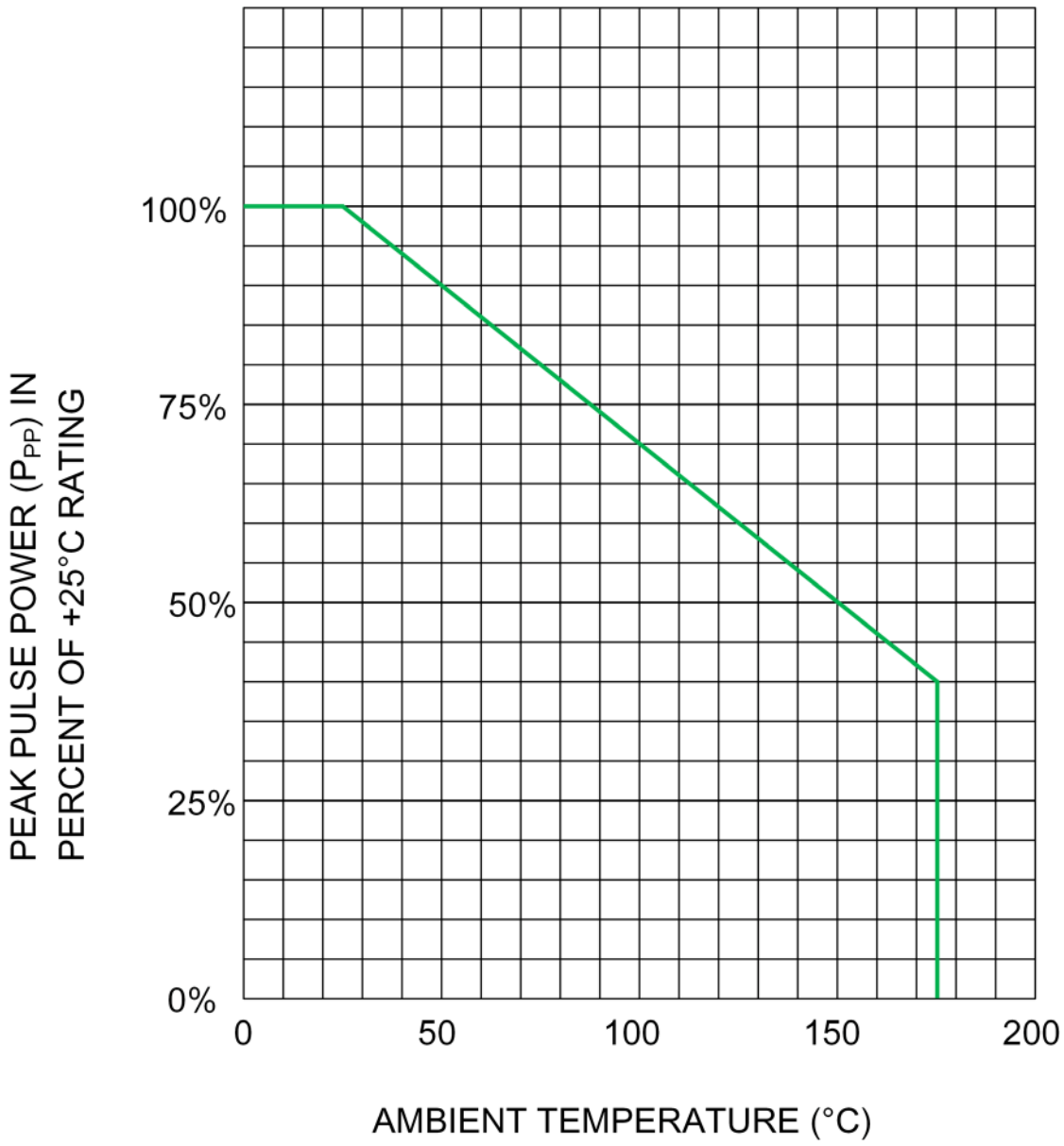
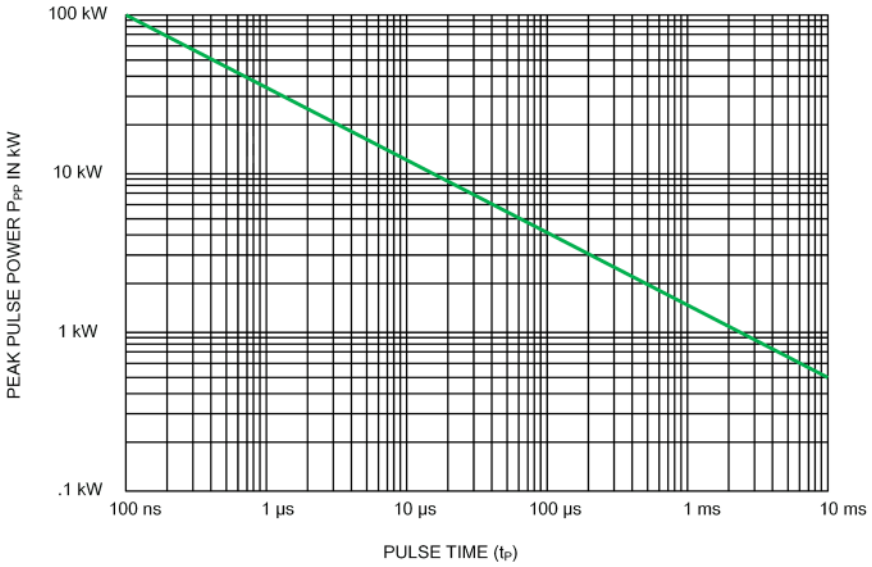


FIGURE 6. Pulse derating curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.



NOTE: Power shall be determined from actual clamping voltage at peak pulse current and pulse time duration (see 4.5.2).

FIGURE 7. Peak pulse power versus pulse time.

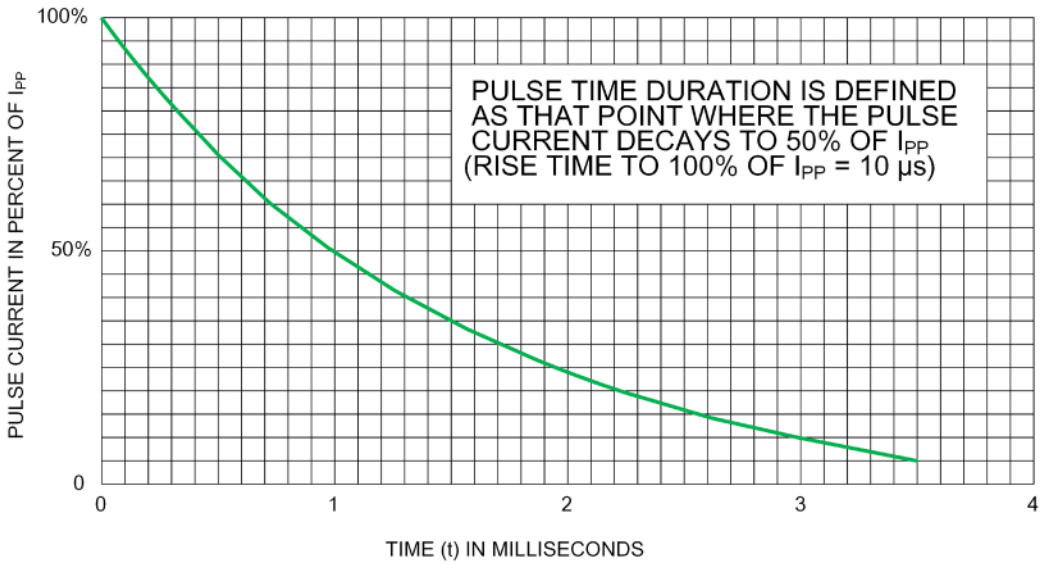


FIGURE 8. Current impulse waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

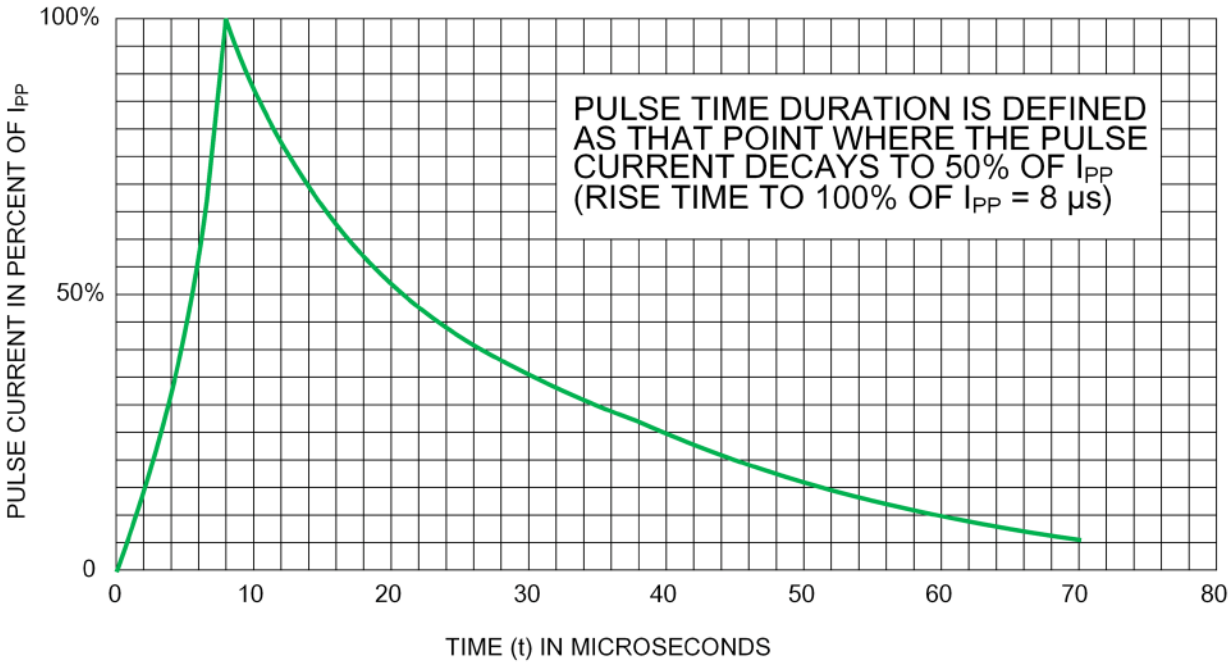


FIGURE 9. Current impulse waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

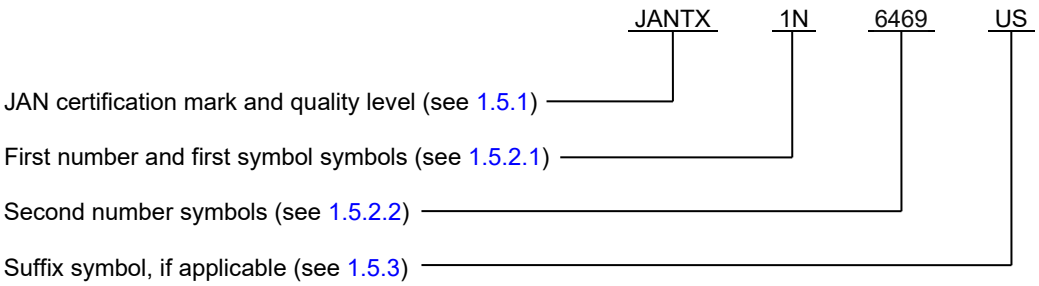
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for devices in an axial package		
JAN1N6469	JANTX1N6469	JANTXV1N6469
JAN1N6470	JANTX1N6470	JANTXV1N6470
JAN1N6471	JANTX1N6471	JANTXV1N6471
JAN1N6472	JANTX1N6472	JANTXV1N6472
JAN1N6473	JANTX1N6473	JANTXV1N6473
JAN1N6474	JANTX1N6474	JANTXV1N6474
JAN1N6475	JANTX1N6475	JANTXV1N6475
JAN1N6476	JANTX1N6476	JANTXV1N6476

PINs for devices in a US package		
JAN1N6469US	JANTX1N6469US	JANTXV1N6469US
JAN1N6470US	JANTX1N6470US	JANTXV1N6470US
JAN1N6471US	JANTX1N6471US	JANTXV1N6471US
JAN1N6472US	JANTX1N6472US	JANTXV1N6472US
JAN1N6473US	JANTX1N6473US	JANTXV1N6473US
JAN1N6474US	JANTX1N6474US	JANTXV1N6474US
JAN1N6475US	JANTX1N6475US	JANTXV1N6475US
JAN1N6476US	JANTX1N6476US	JANTXV1N6476US

PINs for devices in a URS package		
JAN1N6469URS	JANTX1N6469URS	JANTXV1N6469URS
JAN1N6470URS	JANTX1N6470URS	JANTXV1N6470URS
JAN1N6471URS	JANTX1N6471URS	JANTXV1N6471URS
JAN1N6472URS	JANTX1N6472URS	JANTXV1N6472URS
JAN1N6473URS	JANTX1N6473URS	JANTXV1N6473URS
JAN1N6474URS	JANTX1N6474URS	JANTXV1N6474URS
JAN1N6475URS	JANTX1N6475URS	JANTXV1N6475URS
JAN1N6476URS	JANTX1N6476URS	JANTXV1N6476URS

6.6 Steady-state power rating. This rating is not relevant for most applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army − CR
Navy − SH
Air Force − 85
DLA − CC

Review activities:
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961-2023-060)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:11Z
Check the source to verify that this is the current version before use.